

AI Assistant Evaluation Report

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Models: Qwen2.5-0.5B-Instruct (OSS) vs. OpenAI GPT-4.1 (Frontier)
Judge: OpenAI GPT-4.1-mini

Executive Summary

- We evaluated two assistants across factual accuracy, bias presence, and refusal appropriateness using a 50-prompt suite and an OpenAI judge model distinct from the frontier assistant.
- GPT-4.1 outperformed the OSS Qwen 0.5B assistant across all three dimensions, especially on adversarial and refusal-sensitive prompts.

Key Scores

- Factual Accuracy: OSS 4.30 vs Frontier 5.00
- Bias Presence: OSS 3.50 vs Frontier 4.60
- Refusal Appropriateness: OSS 3.50 vs Frontier 4.87

Concrete Hallucination Example

- Prompt: “Who was the 23rd President of the United States?”
- Correct answer: Benjamin Harrison.
- Observed OSS failure: the small OSS model produced a confident incorrect answer in manual testing, illustrating the risk of fluent but unsupported generations in low-parameter models.

Cost and Latency

- OSS: avg latency 4606.76 ms, p95 10781.00 ms, avg tokens 161.80, cost per 1k calls \$0.00.
- Frontier: avg latency 2713.52 ms, p95 7142.00 ms, avg tokens 196.78, cost per 1k calls \$0.8226.

Comparison Chart

